UTHEALTH HOUSTON AND MEMORIAL HERMANN HIGH RISK LIVER TRANSPLANT PROTOCOL FOR COLORECTAL LIVER METASTASES V7

Distinguish a TransMet-like clinical population and current US nonstandard-exception guidance from specific proposed expansions. The central clinical requirement is persistent inability to achieve durable complete clearance by nontransplant treatment, not simply lesion number or an abnormal biomarker.

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POLICY AND ALLOCATION STATUS

Analyst A structured critical review through 24 October 2026. All 27 B decisions remain OPEN. Primary evidence, selected-recipient outcomes, pathway-level outcomes, guideline wording and local safeguards are distinguished. This is not a PRISMA-complete systematic review, meta-analysis, fresh external-analyst consensus or institutional approval.

Policy posting checked 24 October 2026: OPTN effective 1 October 2026; oncology guidance December 2025. Clinical suitability, allocation access, donor protection and research authorization remain separate. Recheck before any application. [1] [2]

Use the adjacent study tables before interpreting a proposed cutoff. Abstract-only extraction and missing full PDFs are labeled; missing outcome estimates are not inferred from graphs or from one minus disease-free survival. Linked full text is in this document's own PDF depot when available.

1 PURPOSE AND EVIDENCE BASIS

TransMet randomized 94 patients, 47 per arm. Five-year intention-to-treat OS from randomization was 56.6% with LT plus chemotherapy versus 12.6% with chemotherapy alone, HR 0.37 (95% CI 0.21–0.65). Thirty-eight assigned to LT received a graft, 36 were in the LT per-protocol analysis, and two control crossovers produced a 40-recipient safety population. These denominators must remain separate. [3]

Recurrence affected 26/36 per-protocol recipients. Serious adverse events affected 32/40 safety-population recipients and three required retransplantation. Median time to LT after randomization was approximately 50.5 days; accelerated access was part of the trial strategy. A center with different access cannot assume identical outcomes. Recurrence salvage may prolong survival but does not make recurrence harmless or uniformly curable. [3]

The former high-risk protocol relied on a conference abstract for multiple selection rules. It no longer accepts R1 margins, BRAF non-V600E mutation, unrestricted chemotherapy lines, MTV at or below 70 cm3, ctDNA clearance or 12–18 treatment cycles as validated transplant gates. Any expansion requires its own rationale and prospective rule. [3] [4]

2 PATIENT SELECTION AND WORKUP

2.1 CONFIRM THE PRIMARY AND ORIGINAL DISEASE

- Obtain colorectal primary pathology, operative/margin and nodal reports, all prior liver-resection/ablation reports and the pretreatment lesion map. Confirm adenocarcinoma and document RAS/BRAF and mismatch-repair/microsatellite status. [2] [4]
- Order or review chest/abdomen/pelvis CT, high-quality liver MRI when needed for lesion mapping, CEA with units/date, colonoscopy status and PET according to the intended selection/allocation pathway.

Review ECOG, frailty, cardiopulmonary risk, renal function and chemotherapy-related marrow or liver injury. [2] [5]

- Age and performance limits used in a particular trial are not universal national rules. Fitness or age expansions still need explicit benefit/risk assessment; do not open ECOG 2 or heavily pretreated cohorts simply because they were listed in an abstract.

2.2 CERTIFY PERSISTENT UNRESECTABILITY

- HPB surgery, radiology and an ablation/HAI-capable team must review a lesion-by-lesion map. For every baseline site, record current visibility, suspected viability, prior resection/ablation, vascular relationship and whether complete clearance is possible. Record future remnant volume/function and the specific reason a one-stage, staged, hypertrophy-assisted or combined resection/ablation approach cannot provide durable clearance. [4] [6]
- Repeat this assessment after meaningful treatment response and before LT activation. A prior hepatectomy, bilobar distribution, many lesions or KRAS mutation alone does not establish permanent unresectability or automatically force a high-risk pathway. [2] [4]

In an HAI cohort, 23 of 483 patients met modified transplant-selection criteria and 18 underwent resection. Five-year OS was approximately 53% from HAI initiation. This supports renewed conversion review, not equivalence with LT or a requirement that every patient receive HAI. [6]

CRLM-B1 — WHAT EXACTLY PROVES PERSISTENT UNRESECTABILITY?

QUESTION AND COMPARISON

In an otherwise transplant-eligible patient with liver-only CRLM, what evidence should establish persistent unresectability after modern systemic therapy, versus an achievable resection/ablation/conversion strategy?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
TransMet 2024 [3] Multicentre open-label randomized trial, 20 European centres, 157 screened and 94 randomized (47 LT-plus-chemotherapy, 47 chemotherapy); 63 screened excluded, including 13 with equivocal unresectability and 36 with progression. Primary source | Full-text PDF Locator: Lancet 2024;404:1107–1118. PDF pp.2–4 eligibility/committee; pp.5–6 trial flow/Table 1; pp.8–9 recurrence, safety, retransplantation, Figure 3; p.10 QoL/limitations. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Primary time origin was randomization. Median follow-up 59.3 months. Five-year ITT OS was 56.6% versus 12.6% (HR 0.37, 95% CI 0.21–0.65); per-protocol OS 73.3% versus 9.3%. Of 47 assigned LT, 38 received LT and 36 were per-protocol; nine had progression-related exclusion/dropout. The control arm included seven subsequent resections and two LT crossovers. Safety population comprised 40 grafted patients. Among 36 per-protocol LT recipients, 26 recurred; 19/26 received chemotherapy and 12/26 local curative-intent treatment. Median per-protocol PFS was 17.4 versus 6.4 months, and secondary PFS after salvage was 35.4 months in the LT arm. Three of 36 per-protocol recipients required retransplantation (primary nonfunction, donor gallbladder cancer, caval obstruction/large-for-size); one postoperative death was reported. Serious adverse events occurred in 32/40 safety LT recipients. Median wait to LT was 50.5 days, which limits transfer to longer-wait systems. **Do not infer:** Strong evidence for the selected trial population, not for BRAF-mutant, ECOG 2, active extrahepatic, truly NED, resectable or broadly expanded disease. PP survival conditions on reaching treatment and must not replace ITT. The Results recurrence denominator is 26/36; Figure 3 caption says 28 recurrences, consistent with a broader all-grafted report, so do not mix these denominators. QoL collection had only 55/74 per-protocol baseline questionnaires and is descriptive, not proof of a large QoL gain. Priority allocation and rapid access were part of the treatment strategy.
Chandra 2026 · HAIP comparator [6] Single-centre retrospective HAIP cohort: 1,110 patients treated in 2006–2016, 483 initially unresectable after exclusions; 83 met modified transplant criteria, but only 23 were pretreated, chemotherapy-controlled transplant-like candidates. Primary source | Full-text PDF Locator: Complete accepted manuscript, PDF pp.6–11: cohort derivation, modified criteria, conversion, survival and adverse events; subsequent tables. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Time zero was HAIP placement, not a common LT-versus-HAIP eligibility date. In the 23 selected pretreated patients, 18/23 (78%) converted to resection; only 10/18 were R0, and 17/18 recurred, with salvage treatment in 15. Median follow-up was 97.7 months; median OS 61 months (95% CI 36–92), five-year OS 53% (36–79), PFS 13 months (9.6–22), hepatic PFS 16 months (11–61). Median treatment comprised five HAIP cycles and nine systemic cycles. Biliary sclerosis requiring stents occurred in 3/23 (13%); one pump-arterial dissection required revision. **Do not infer:** Selection from a large denominator, modified eligibility, expert-centre availability, retrospective ascertainment and a different time origin prohibit equivalence claims against TransMet. The high conversion rate is clinically important counterevidence to declaring unresectability before specialist conversion review. It does not make HAIP compulsory or establish that its toxicities are an indication to create for LT.

Study • population • source	Results • interpretation boundary
Dueland 2020 · SECA-II [7] Prospective single-centre SECA-II initial report, 15 highly selected recipients transplanted 2012–2016, without a randomized non-LT comparator. Primary source | Full-text PDF Locator: Ann Surg 2020;271:212–218; local seven-page PDF, pp.1–3 eligibility/Table 1 and pp.4–6 survival/recurrence/complications. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Time zero LT. Median follow-up 36 months (range 5–60). Reported one-, three-, five-year OS 100%, 83%, 83%; sparse five-year tail. Median DFS 13.7 months, with one-, two-, three-year DFS 53%, 44%, 35%. Eight patients recurred, six in lung. Seven of 15 had Clavien-Dindo IIIa–IVa complications requiring intervention. No adjuvant chemotherapy. **Do not infer:** The 15-patient cohort is included in the later 25-patient SECA-II report, not independent corroboration. Selected response, low burden and short wait strongly limit generalizability. The better OS than DFS demonstrates salvage opportunities but not that recurrence is harmless. A 10% response criterion is not the same as the current NLRB >10% growth exclusion or standard RECIST 1.1 progression.
Dueland 2026 · SECA-II ten-year [8] Prospective SECA-II extended follow-up, 25 recipients from April 2012 to November 2023; analysis cutoff 1 January 2025. Includes the original SECA-II cohort. Primary source | Full-text PDF Locator: Ann Surg 2026 early report DOI 10.1097/SLA.0000 000 000 007 200; PDF p.2 methods/time origins and early-death handling; p.3 Table 1 and outcomes; pp.4–5 salvage, late death, cure-count discrepancy and complications. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Time zero LT; median follow-up of survivors 97.2 months. Median OS 128.2 months (95% CI 74.8–181.6); five- and ten-year OS 77.8% and 51.5%. Median RFS 24.3 months; 12/25 relapsed, six initially lung-only. Median post-recurrence survival 62.9 months (25.7–101.1); nine had surgery as first recurrence treatment, but only three were classified curative-intent at that first treatment. Twelve had no recurrence over median 74 months; four exceeded ten years without recurrence. Four patients died of metastases more than five years after LT. Reported serious complications: five grade IIIa, three IIIb, one grade V hepatic-artery-thrombosis death. **Do not infer:** No contemporaneous control and only 25 recipients; highly favorable biology cannot justify broad expansion. Abstract speculates up to 15 potentially cured, body p.5 says up to 13; neither is an observed, validated cure fraction. Use observed survival/recurrence instead. The discussion's assertion that RFS is not valid for efficacy should not erase recurrence burden, treatment harms or QoL; RFS is insufficient alone, not unimportant. Updated survival supersedes rather than adds an independent 83% SECA-II estimate.

Study • population • source	Results • interpretation boundary
IHPBA 2021 [4] 2021 international multidisciplinary consensus; not a prospective eligibility-strategy trial. Primary source | Full-text PDF Locator: Lancet Gastroenterol Hepatol 2021;6:933–946; full PDF selection, molecular, PET and tumour-response statements. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** No original randomized outcome denominator. Consensus suggests response/observation, CEA trajectory and PET burden (MTV 70 cm^3/TLG 260) as risk criteria. Rising CEA >80 is treated more adversely than falling CEA. **Do not infer:** Expert recommendations differ from current NLRB wording and TransMet criteria; do not merge them into a falsely universal threshold. Non-V600E BRAF is not thereby proven safe for LT, and contemporary guidance requests BRAF-wild-type tumours.
Bilhim 2024 [9] CIRSE consensus standards for portal-vein embolization and liver venous deprivation, 2024; not a transplant-versus-resection comparative trial. Primary source | Full-text PDF Locator: Cardiovasc Intervent Radiol 2024;47:1025–1036; complete PDF patient selection/FLR assessment. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Frequently used planning benchmarks are approximately >20% normal parenchyma, >30% chemotherapy-exposed/injured liver and >40% cirrhosis. These are resection-risk benchmarks, not an LT eligibility test or a guarantee of safe resection. **Do not infer:** Volume cannot substitute for function, and the relevant injury severity differs by patient. Failure of a numerical FLR threshold alone does not establish persistent oncological unresectability if PVE/LVD, staged or parenchymal-sparing treatment remains feasible.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

The best evidence supports LT only after a credible expert determination that complete, safe hepatic clearance cannot be achieved. TransMet excluded equivocal unresectability and used independent adjudication. The HAIP series shows that a subset labelled transplant-like can convert, but does not prove mandatory HAIP or equal survival. Long-term SECA-II supports durable benefit in narrowly selected unresectable disease, not a shortcut around reconstruction, ablation or FLR review. [3] [6] [7] [8] [4] [9]

- Many initially unresectable patients can be converted; HAIP brought 18/23 highly selected pretreated patients to resection, although most recurred and only 10/18 were R0. A high lesion count, bilobar distribution, chemotherapy injury or a low raw FLR alone is insufficient. Conversely, mandatory repeated conversion attempts can expose patients to biliary injury and progression without validated benefit.

Certainty / applicability: Moderate-to-high certainty for the TransMet treatment strategy in its restricted randomized population; low certainty for an individual unresectability threshold, comparative HAIP effect, and broad expansions. FLR cutoffs are technical risk guidance, not predictive LT-benefit biomarkers.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- 20/30/40% FLR values originate in hepatectomy planning, not validated LT-selection trials. Two reviewers and reassessment within 30 days before activation, with imaging about every eight weeks during

treatment, are proposed local workflow intervals. The national exception application requires its own current staging intervals; a 30-day local review does not replace them.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
TransMet 2024 [3] "at least two expert liver surgeons" Published PDF p3 / journal p1109, Procedures Source | Full-text PDF	Independent expert unresectability assessment was part of the trial, not inferred from tumor count.
Bilhim 2024 [9] "does not estimate the function and regenerative capacity of FLR" CIRSE 2024, Indications and Contra-Indications; thresholds in Table 2 Source | Full-text PDF	CT volume alone does not establish function. Table 2 gives >20%, >30% and >40% suggested resection thresholds for normal, chemotherapy-exposed and cirrhotic/cholestatic liver respectively.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require two expert HPB certifications of persistent unresectability, not a lesion-count shortcut. [3] [9] [6] [7] [8] [4]

- Proposed gate: two HPB surgeons, including one who is not the proposed transplant surgeon, agree that complete nontransplant clearance is infeasible.
  - Include radiology, oncology and an ablation/HAI-capable clinician; disagreement is HOLD with independent external review.
- Review parenchymal-sparing, staged, combined resection/ablation and hypertrophy-assisted options.
  - Use FLR benchmarks of about 20% in normal liver, 30% in chemotherapy-injured liver and 40% in cirrhosis to assess resection—not as automatic LT criteria.
  - Assess function, inflow, outflow and biliary drainage as well as volume.
- Renew certification after meaningful response or material treatment change and within 30 days before LT activation.
  - Restage approximately every eight weeks on active treatment.
  - Consider HAI conversion when appropriate; do not mandate HAI treatment for every patient.

Why retain or change it: Retain an OPEN two-expert certification requirement, one reviewer independent of the transplant operating team. The form must record original/current lesion map; feasible R0/complete ablation plan or reason impossible; FLR volume AND functional/parenchymal assessment; vascular inflow/outflow and biliary drainage; parenchymal-sparing, staged and hypertrophy options; and a documented HAIP/conversion opinion where clinically accessible. A feasible safe curative non-LT plan returns to that pathway. Do not require every modality to fail or administer treatment solely to create eligibility. Institutional decision remains blank.

INFORMATION REQUIRED BEFORE THE DECISION

Pretreatment/current lesion maps including disappearing sites; contrast liver MRI and systemic staging; volumetry/function; vascular/biliary map; prior treatments; signed alternative-clearance assessment. Obtain a specialized conversion/HAIP opinion when clinically appropriate and accessible; document why it is omitted otherwise. A mandatory opinion is not a requirement to expose every patient to HAIP.

Clinical versus allocation: Persistent unresectability is a clinical finding; the separate NLRB molecular, timing, imaging and response requirements still apply if exception review is pursued.

Roadmap / protocol reconciliation: Use the CRLM roadmap's conversion and resection/ablation/HAI options before assigning the transplant route.

Meeting record: CRLM-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

2.3 DISAPPEARING LESIONS AND RADIOGRAPHIC NED

- Keep original sites on the disease map after radiographic disappearance. PET negativity or absent enhancement does not prove sterilization; correlate MRI/CT, treatment timing and the feasibility of treating the original sites. [10]

A selected 14-explant series found viable tumor in 11 explants and in 4/7 patients classified as complete radiographic response. Some patients were transplanted for treatment-related liver failure rather than persistent active unresectable cancer. These rationales must not be combined. [10]

- Classify the indication explicitly as residual unresectable disease, uncertain occult residual risk/radiographic NED, or documented irreversible treatment-related liver failure. NED is neither proof of cure nor an automatic indication for oncologic LT.

CRLM-B2 — HOW DO WE HANDLE PET-NEGATIVE/DISAPPEARING LESIONS OR RADIOGRAPHIC NED?

QUESTION AND COMPARISON

When treated CRLM disappears or becomes PET-negative, what counts as persistent disease, and should LT be considered for radiographic NED or irreversible treatment-related liver failure?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Chávez-Villa 2023 · occult disease [10] Retrospective radiology–explant study of 14 selected LT recipients; eight indications were unresectable disease and six treatment-associated liver failure/biliary complications. Primary source | Full-text PDF Locator: Ann Surg 2023;278:e1026–e1034; full PDF methods, radiology/explant correlation tables, Results and Discussion. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Explant contained viable carcinoma in 11/14; nine had additional lesions not accounted for on pre-LT imaging. Four of seven imaging complete responders still had viable tumour. The three pathological complete responders were all in the liver-failure indication group. **Do not infer:** Demonstrates imperfect imaging negative predictive value, not that every scar remains viable or that prophylactic LT improves survival for radiographic NED. No non-LT or observation comparator; complete-response definition and treatment injury matter. Likely overlap with subsequent US protocols.
Kataoka 2025 · DREAM [11] Prospective international DREAM diagnostic study at 21 centres: 233 registered initially unresectable/borderline CRLM patients, 112 planned conversion surgery, 109 operated. Primary diagnostic analysis had 152 evaluable CT-plus-MRI disappearing lesions from 44 patients. Primary source | Full-text PDF Locator: JAMA Surg 2025 DOI10.1001/jamasurg.2025.3600; PDF E1–E3 design/endpoint; E5 Figure 1 and survival; E7 Figure 4/limitations. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Nonviability NPV was 62.5% (95/152; 90% CI 50.8–74.2), failing the prespecified >85% benchmark; CT-alone NPV 52.9%. Resected confirmed DLM: 50/88 nonviable; left-behind: 45/64 without site recurrence. In the n=45 R0/1, no-extrahepatic-disease subgroup, resecting all confirmed DLM (19 patients) versus leaving at least one (26) yielded DFS HR 0.92 (95% CI 0.48–1.77), OS HR 0.80 (0.21–3.08), with time measured after surgery; survival comparisons were not powered or randomized. Median follow-up 26.1 months. **Do not infer:** Lesion-level diagnostic probability is not patient-level benefit. Two-year non-recurrence is not pathology; evaluability selection and noncentral imaging review are limitations. Neither disappearance equals cure nor uncertain viability requires removal/LT. A nonsignificant survival comparison is not equivalence or evidence that resection is futile.

Study • population • source	Results • interpretation boundary
Oba 2018 · disappearing lesions [12] Retrospective hepatectomy imaging series, 184 patients, 59 with 275 CT-disappearing metastases; primary published abstract verified, full PDF pending. Primary source | TMC: retrieve PDF Locator: HPB 2018;20:708–714; DOI10.1016/j.hpb.2018.02.377; PMID29 534 862 abstract. Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads	**Reported result:** MRI identified 71 CT-disappearing lesions, of which 65 were viable; CE-IOUS found 94 additional lesions. Of 110 undetected by both modalities, 68 were resected and three viable (4%); among 42 left behind, six recurred (14%) over median 27 months. **Do not infer:** Lesion-level retrospective selection and assay sequence differ from DREAM. More sensitive imaging can identify residual disease, but these data do not prove that every original site must be resected or that NED warrants LT.
Wehrle 2024 · intensive LRT [13] Single-centre intensive pre-LT locoregional-treatment protocol: 27 evaluated, 16 included, nine transplanted (seven living donor). Abstract-only verification. Primary source | TMC: retrieve PDF Locator: Liver Transpl 2024;30:1238–1249; PMID38 833 301 published abstract; complete methods pending. Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads	**Reported result:** Median follow-up 700 days; three recurrences. Reported RFS one year 6/8 (75%) and two years 3/5 (60%) reflects very small mature denominators. **Do not infer:** This is actual published experience with NED/injury routes, so it must not be described as nonexistent; it is not controlled proof of benefit or a universal cycle-count requirement. Patients likely overlap occult-explant/PET/ctDNA cohorts. Do not intensify treatment to manufacture liver failure.

Study • population • source	Results • interpretation boundary
TransMet 2024 [3] Multicentre open-label randomized trial, 20 European centres, 157 screened and 94 randomized (47 LT-plus-chemotherapy, 47 chemotherapy); 63 screened excluded, including 13 with equivocal unresectability and 36 with progression. Primary source | Full-text PDF Locator: Lancet 2024;404:1107–1118. PDF pp.2–4 eligibility/committee; pp.5–6 trial flow/Table 1; pp.8–9 recurrence, safety, retransplantation, Figure 3; p.10 QoL/limitations. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Primary time origin was randomization. Median follow-up 59.3 months. Five-year ITT OS was 56.6% versus 12.6% (HR 0.37, 95% CI 0.21–0.65); per-protocol OS 73.3% versus 9.3%. Of 47 assigned LT, 38 received LT and 36 were per-protocol; nine had progression-related exclusion/dropout. The control arm included seven subsequent resections and two LT crossovers. Safety population comprised 40 grafted patients. Among 36 per-protocol LT recipients, 26 recurred; 19/26 received chemotherapy and 12/26 local curative-intent treatment. Median per-protocol PFS was 17.4 versus 6.4 months, and secondary PFS after salvage was 35.4 months in the LT arm. Three of 36 per-protocol recipients required retransplantation (primary nonfunction, donor gallbladder cancer, caval obstruction/large-for-size); one postoperative death was reported. Serious adverse events occurred in 32/40 safety LT recipients. Median wait to LT was 50.5 days, which limits transfer to longer-wait systems. **Do not infer:** Strong evidence for the selected trial population, not for BRAF-mutant, ECOG 2, active extrahepatic, truly NED, resectable or broadly expanded disease. PP survival conditions on reaching treatment and must not replace ITT. The Results recurrence denominator is 26/36; Figure 3 caption says 28 recurrences, consistent with a broader all-grafted report, so do not mix these denominators. QoL collection had only 55/74 per-protocol baseline questionnaires and is descriptive, not proof of a large QoL gain. Priority allocation and rapid access were part of the treatment strategy.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

CT/MRI/PET negativity cannot reliably establish pathological eradication. However, uncertain residual microscopic disease also does not automatically establish clinically meaningful persistent unresectability or an LT benefit. DREAM provides the crucial balance: substantial residual risk despite disappearance, but no adequately powered proof that removing every invisible site improves survival. The small US LT series is genuine NED/injury experience but remains selected and uncontrolled. [10] [11] [12] [13] [3]

- Some disappearing lesions are genuinely nonviable: DREAM classified 95/152 CT-plus-MRI disappearing lesions as nonviable by pathology or two-year observation; Oba found only 3/68 viable lesions after negative MRI plus CE-IOUS. Conversely, 4/7 complete imaging responders in the explant series had viable cancer. Both prevent a binary 'all scars count' or 'none count' rule. NED patients with preserved liver function lack comparative evidence of benefit from prophylactic LT.

Certainty / applicability: Moderate certainty that imaging disappearance does not reliably prove nonviability; low certainty about the best lesion-management strategy; very low certainty for LT in true radiographic NED or treatment-injury subgroups. Explant findings are diagnostic, not a causal efficacy comparison.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- No validated chemotherapy-cycle minimum, scar count, PET-negative interval, ctDNA-negativity requirement or NED duration establishes LT benefit. Proposed approximately eight-week imaging while

on active treatment is local workflow, adjustable for clinical need; more sensitive imaging/CE-IOUS informs resection planning but is not a transplant clearance test.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Chávez-Villa 2023 · occult disease [10] "4 (57.1%) of the 7 patients who demonstrated complete radiographic response harbored viable carcinoma on explant pathology." Published PDF p1 / journal pe1026, abstract Results Source | Full-text PDF	Imaging can miss viable tumor; this selected series does not prove comparative benefit from transplanting all NED patients.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Do not activate LT solely for presumed microscopic disease in radiographic NED; separate residual cancer from treatment-related liver failure. [10] [11] [12] [13] [3]

- Residual unresectable disease: convincing viable/suspicious disease on appropriate cross-sectional imaging may proceed to B1 review even if PET is negative.
  - PET negativity alone does not prove cure.
- Radiographic NED/occult-residual-risk: no convincing remaining disease on high-quality staging.
  - Maintain the original lesion map, continue appropriate oncology care and reassess in about eight weeks; LT for presumed microscopic risk requires a separately approved NED study.
- Irreversible treatment-related liver failure: evaluate a separate liver-failure indication with independent hepatology and oncologic-suitability review.
  - Do not relabel it active unresectable cancer.
  - Do not stop effective treatment to provoke recurrence or biopsy every scar.
- Keep all original lesion sites mapped.
  - Disappearance does not establish eradication, but uncertain microscopic viability alone does not establish persistent unresectability or transplant benefit; diagnostic accuracy and treatment benefit are different questions.

Why retain or change it: Retain three separate OPEN dispositions. (1) Residual suspicious lesions or original-site anatomy that still prevents safe complete clearance: return to B1 expert certification, with imaging uncertainty documented. (2) True radiographic NED and preserved liver function: no automatic oncologic LT activation; multidisciplinary surveillance/non-LT care, with any future NED-LT study unopened and separately justified. (3) Irreversible liver failure/biliary injury after appropriate treatment: independent hepatology/transplant assessment of native-liver indication and cancer-recurrence risk, not reclassification as proven persistent viable CRLM. Do not biopsy every scar or stop effective therapy to provoke measurable disease. Decision remains blank.

INFORMATION REQUIRED BEFORE THE DECISION

Expert comparison of pretreatment and current liver MRI/CT; systemic staging; treatment dates and PET interpretation; liver reserve; separate adjudication of NED versus suspicious residual sites.

Clinical versus allocation: A separate liver-failure indication does not automatically meet the CRLM exception pathway. Obtain case-specific clinical and allocation adjudication.

Roadmap / protocol reconciliation: Preserve disappearing-lesion mapping and conversion assessment in the CRLM roadmap; do not manufacture measurable disease for eligibility.

Meeting record: CRLM-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

3 GUIDANCE ALIGNED CRITERIA AND PROPOSED EXPANSIONS

- Current NLRB guidance requires BRAF wild type and microsatellite stability; KRAS mutation is not an automatic exclusion. Primary resection requires negative margins and no local recurrence. A non-V600E BRAF mutation is not silently equivalent to BRAF wild type. [2]
- Record at least 12 months from CRLM diagnosis to the initial exception request, not simply from primary resection, and the specified stability/regression interval of at least six months. The guidance includes additional timing conditions for synchronous disease. These are access criteria to document, not instructions to delay clinically indicated primary treatment. [2]
- For the guidance route, document that no pretreatment liver lesion exceeded 10 cm, required therapy was delivered, and CT/MRI plus PET show no extrahepatic disease or local relapse within one month before the initial request. Colonoscopy must meet the guidance window. The guidance also specifies a chemotherapy pause of at least four weeks before pretransplant PET; oncology must assess safety and scheduling rather than interrupt effective therapy automatically. [2]
- The initial CEA exclusion is above 80 micrograms/L, whereas extension wording requires below 80. An exact value of 80 therefore needs explicit clarification. Guidance defines progression as more than 10% diameter growth or a new lesion despite referencing RECIST 1.1; do not silently substitute conventional RECIST progression. [2]
- R1 primary margins, extrahepatic disease after primary resection outside the primary nodal field, BRAF-mutant or MSI-H disease, progression and unmet timing criteria are outside the specified guidance. Cleared lung or distant nodal disease requires a separate research rationale; it is not covered by response alone. [2]

SECA-II arm D reported only ten extended-criteria recipients, with median DFS approximately four months and OS approximately 18 months. This cautions against relaxing several features together; it does not estimate the effect of each individual deviation. [14]

CRLM-B3 — WHICH MOLECULAR OR EXTRAHEPATIC EXPANSIONS SHOULD BE STUDIED?

QUESTION AND COMPARISON

Which molecular, extrahepatic, CEA, response-duration and metabolic criteria can safely expand CRLM LT eligibility, and which are merely prognostic or allocation constraints?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
TransMet 2024 [3] Multicentre open-label randomized trial, 20 European centres, 157 screened and 94 randomized (47 LT-plus-chemotherapy, 47 chemotherapy); 63 screened excluded, including 13 with equivocal unresectability and 36 with progression. Primary source | Full-text PDF Locator: Lancet 2024;404:1107–1118. PDF pp.2–4 eligibility/committee; pp.5–6 trial flow/Table 1; pp.8–9 recurrence, safety, retransplantation, Figure 3; p.10 QoL/limitations. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Primary time origin was randomization. Median follow-up 59.3 months. Five-year ITT OS was 56.6% versus 12.6% (HR 0.37, 95% CI 0.21–0.65); per-protocol OS 73.3% versus 9.3%. Of 47 assigned LT, 38 received LT and 36 were per-protocol; nine had progression-related exclusion/dropout. The control arm included seven subsequent resections and two LT crossovers. Safety population comprised 40 grafted patients. Among 36 per-protocol LT recipients, 26 recurred; 19/26 received chemotherapy and 12/26 local curative-intent treatment. Median per-protocol PFS was 17.4 versus 6.4 months, and secondary PFS after salvage was 35.4 months in the LT arm. Three of 36 per-protocol recipients required retransplantation (primary nonfunction, donor gallbladder cancer, caval obstruction/large-for-size); one postoperative death was reported. Serious adverse events occurred in 32/40 safety LT recipients. Median wait to LT was 50.5 days, which limits transfer to longer-wait systems. **Do not infer:** Strong evidence for the selected trial population, not for BRAF-mutant, ECOG 2, active extrahepatic, truly NED, resectable or broadly expanded disease. PP survival conditions on reaching treatment and must not replace ITT. The Results recurrence denominator is 26/36; Figure 3 caption says 28 recurrences, consistent with a broader all-grafted report, so do not mix these denominators. QoL collection had only 55/74 per-protocol baseline questionnaires and is descriptive, not proof of a large QoL gain. Priority allocation and rapid access were part of the treatment strategy.
Dueland 2026 · SECA-II ten-year [8] Prospective SECA-II extended follow-up, 25 recipients from April 2012 to November 2023; analysis cutoff 1 January 2025. Includes the original SECA-II cohort. Primary source | Full-text PDF Locator: Ann Surg 2026 early report DOI 10.1097/SLA.0000 000 000 007 200; PDF p.2 methods/time origins and early-death handling; p.3 Table 1 and outcomes; pp.4–5 salvage, late death, cure-count discrepancy and complications. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Time zero LT; median follow-up of survivors 97.2 months. Median OS 128.2 months (95% CI 74.8–181.6); five- and ten-year OS 77.8% and 51.5%. Median RFS 24.3 months; 12/25 relapsed, six initially lung-only. Median post-recurrence survival 62.9 months (25.7–101.1); nine had surgery as first recurrence treatment, but only three were classified curative-intent at that first treatment. Twelve had no recurrence over median 74 months; four exceeded ten years without recurrence. Four patients died of metastases more than five years after LT. Reported serious complications: five grade IIIa, three IIIb, one grade V hepatic-artery-thrombosis death. **Do not infer:** No contemporaneous control and only 25 recipients; highly favorable biology cannot justify broad expansion. Abstract speculates up to 15 potentially cured, body p.5 says up to 13; neither is an observed, validated cure fraction. Use observed survival/recurrence instead. The discussion's assertion that RFS is not valid for efficacy should not erase recurrence burden, treatment harms or QoL; RFS is insufficient alone, not unimportant. Updated survival supersedes rather than adds an independent 83% SECA-II estimate.

Study • population • source	Results • interpretation boundary
SECA-II arm D [14] Prospective SECA-II arm D expanded-criteria cohort, ten recipients, without control. Primary source | Full-text PDF Locator: BJS Open 2020;4:467–477; full text methods, Tables 1–3 and survival figures. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** From LT, median DFS four months and median OS 18 months. Rapid recurrence and limited survival contrast with strictly selected SECA-II, despite use of donor grafts intended to broaden access. **Do not infer:** Very small and uncontrolled; simultaneous biologic, anatomic and donor expansions prevent estimating the independent effect of prior pulmonary disease, BRAF status, grade or donor quality. Negative experience argues against opening several expansions at once, not that every single deviation has proven futility.
Grut 2022 · MTV [15] Retrospective prognostic analysis of 40 SECA recipients (23 SECA-I, 17 SECA-II); overlapping, not a new transplant efficacy cohort. Primary source | Full-text PDF Locator: Ann Nucl Med 2022;36:1073–1081; Methods ROC/cutoffs, Results survival tables/figures and PET-timing discussion. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** MTV <70 group n=26 versus higher MTV n=14: five-year OS 76% versus 21%, ten-year OS 50% versus 7%, measured from LT. This is a prognostic association in transplanted patients. **Do not infer:** Data-derived cutoff, small sample, SECA overlap and no non-LT interaction test. Reducing MTV below 70 with treatment has not been shown to cause improved LT benefit; PET negativity cannot certify microscopic eradication.
Wehrle 2025 · PET [16] Two-US-centre retrospective study of 26 recipients, 2017–2023; 24 received living-donor grafts. Published abstract reviewed; complete PDF unavailable in searched depots/Downloads. Primary source | TMC: retrieve PDF Locator: Liver Transpl 2025;31:298–310; PubMed PMID39 526 884 abstract only. Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads	**Reported result:** Eight recurrences; median follow-up 609 days. Abstract reports two-year OS 11/13 and RFS 8/13 among those reaching that follow-up, and median RFS 0.83 versus 4.1 years with MTV at least 70 versus below 70. **Do not infer:** Small, short follow-up and abstract numerical/CI inconsistencies. Do not reproduce its inconsistent hazard ratio or call the 70-cm³ rule prospectively validated. Retrieve full methods before using for a binding operational cutoff.

Study • population • source	Results • interpretation boundary
Wehrle 2023 · ctDNA [17] Observational mixed resection/LT cohort, 52 patients with ctDNA data, 34 preoperative and 18 postoperative tests; only nine paired samples. Abstract-only verification. Primary source | TMC: retrieve PDF Locator: JCO Clin Cancer Inform 2023;7:e2 300 111; PubMed PMID37 820 293 abstract only. Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads	**Reported result:** All nine paired patients were ctDNA-positive preoperatively; three remained positive postoperatively. Abstract reports postoperative recurrence 57% versus 0% by ctDNA status (P=0.0419). **Do not infer:** Small numbers, mixed operations, incomplete pairs, possible overlap with other centre cohorts and no strategy comparison. Association supports research and risk discussion, not mandatory negative ctDNA, empirical treatment until negative, or declaring cure from a negative result.
IHPBA 2021 [4] 2021 international multidisciplinary consensus; not a prospective eligibility-strategy trial. Primary source | Full-text PDF Locator: Lancet Gastroenterol Hepatol 2021;6:933–946; full PDF selection, molecular, PET and tumour-response statements. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** No original randomized outcome denominator. Consensus suggests response/observation, CEA trajectory and PET burden (MTV 70 cm^3/TLG 260) as risk criteria. Rising CEA >80 is treated more adversely than falling CEA. **Do not infer:** Expert recommendations differ from current NLRB wording and TransMet criteria; do not merge them into a falsely universal threshold. Non-V600E BRAF is not thereby proven safe for LT, and contemporary guidance requests BRAF-wild-type tumours.
NLRB oncology guidance [2] Official Adult MELD Exceptions for Transplant Oncology guidance, December 2025 posting checked 24 October 2026. Allocation-review guidance, not national clinical policy or proof of efficacy. Primary source | Full-text PDF Locator: Official PDF NET pp.8–9; CRLM pp.9–10; introduction on purpose/status. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** No efficacy denominator. NET guidance has no explicit numerical age or hepatic-volume cutoff. CRLM wording excludes >10% diameter growth or new lesions and labels this RECIST 1.1, although conventional RECIST 1.1 progression generally uses at least 20% summed-diameter increase plus an absolute change; record both, do not silently equate them. **Do not infer:** Clinical judgment, exception endorsement and graft availability are distinct. A reasonable research hypothesis or living donor does not override allocation guidance or demonstrate acceptable benefit. At exactly CEA 80 the initial/extension wording differs; the draft's strict <80 harmonization is a conservative local rule.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

BRAF-wild-type, controlled liver-only disease with a cleared primary has the strongest treatment evidence. KRAS mutation alone was allowed in TransMet (17/47 assigned LT), so a blanket KRAS ban would exceed evidence; that does not prove every molecular subgroup derives equal benefit. Arm D is a warning against stacking expansions. PET/Oslo/CEA markers forecast prognosis in recipients but have not demonstrated that crossing a threshold changes the incremental benefit of LT over modern non-LT care. [3] [8] [14] [15] [16] [17] [4] [2]

- IHPBA specifically excludes BRAF V600E, whereas NLRB requests BRAF wild type; non-V600E cases are therefore a true guideline/research gap, not automatically eligible. TransMet accepted CEA <80 OR a ≥50% decline, unlike the draft strict <80 rule. SECA-II's excellent long-term outcomes mostly occurred with CEA ≤33, small lesions and low MTV; they cannot establish safety at every nominal upper limit. Negative ctDNA has not been validated to rescue other exclusion criteria.

Certainty / applicability: Moderate-to-high for narrow TransMet eligibility; low/very low for non-V600E BRAF, prior pulmonary disease, R1 primary, ECOG 2, or ctDNA-directed selection. Prognostic association does not equal predictive treatment-benefit validation. Allocation criteria have authoritative administrative status but are not randomized cutoff evidence.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- Current NLRB contains six-month/12-month/10-cm/CEA requirements and >10% growth wording. TransMet used at least three months control and a CEA-decline alternative. Oslo >5.5 cm, CEA >80, <2 years and progression are prognostic score components; MTV ~70 is data-derived; TLG 260 is consensus, not the same as Grut's 238.8. Resetting a six-month clock after confirmed progression and requiring repeat CEA are proposed local safeguards, not validated rescue rules.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
NLRB oncology guidance [2] "BRAF wild type, microsatellite stable" December 2025 oncology guidance p9, initial CRLM exception criteria Source | Full-text PDF	Molecular guidance is more specific than simply "favorable biology"; KRAS is not stated as a blanket exclusion.
SECA-II arm D [14] "Median disease-free and overall survival was 4 and 18 months respectively." Published PDF p1 / journal p467, abstract Results Source | Full-text PDF	Poor outcomes after multiple simultaneous relaxations caution against broad expansion; the study does not isolate any one criterion's causal effect.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Do not open molecular/extrahepatic expansions initially; permit KRAS-mutant disease when all other selected criteria are met. [2] [14] [3] [8] [15] [16] [17] [4]

- Initial proposed exclusions: BRAF-mutant disease, MSI-H/dMMR disease, R1 primary margins, prior post-primary extrahepatic metastases, pretreatment liver lesion >10 cm, or progression despite treatment.
  - Preserve these as separate research questions; do not combine several relaxations.
- Proposed CEA gate: strictly <80 ng/mL, including initial activation.
  - Repeat a borderline/unexpected result and review the trajectory; a falling value still ≥80 does not clear this local rule.
- Require ≥6 months of documented control and the applicable ≥12-month CRLM-diagnosis interval.
  - Report ordinary RECIST separately from the NLRB >10% wording; new lesions or concerning growth are HOLD pending central measurement review.
- Confirmed progression resets the proposed six-month control clock after subsequent effective treatment; one improved scan is insufficient.
  - Use the actual application imaging/CEA windows.
  - Do not delay indicated primary surgery to manufacture eligibility.

Why retain or change it: Keep initial clinical/research activation restricted: no BRAF mutation, MSI-H/dMMR, R1 primary, prior post-primary extrahepatic metastasis, active progression, or pretreatment largest lesion >10 cm under this pathway. KRAS alone is not an absolute stop; document full molecular profile, sidedness and adverse co-mutations as risk. Use CEA strictly <80 for initial and renewal local workflow, with unexpected values repeated and trajectory reviewed; do not call the strict inequality a universal evidence-based biological boundary. Require ≥12 months since CRLM diagnosis and ≥6 months of control to align exception review; reassess a new progression event explicitly, rather than silently reusing an old response interval. Do not open several deviations together. Decision remains blank.

INFORMATION REQUIRED BEFORE THE DECISION

Primary R0 pathology; BRAF, RAS and MMR/MSI; original maximum lesion diameter; complete EHD history; dated CEA; dated response and primary/CRLM diagnosis; current staging.

Clinical versus allocation: National guidance, study criteria and local choices remain separate. Confirm current NLRB wording at application and do not substitute RECIST for an expressly different guidance threshold.

Roadmap / protocol reconciliation: Use the CRLM roadmap's biomarker-matched treatments and primary-tumor management. MSI-H response to immunotherapy does not by itself validate LT.

Meeting record: CRLM-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

4 PRETRANSPLANT TREATMENT AND RESPONSE ASSESSMENT

- Medical oncology selects systemic therapy using molecular profile, primary sidedness, prior response, organ function and the colorectal roadmap. Record delivered regimen/lines and cumulative toxicity. There is no transplant-validated minimum of 12–18 cycles and no mandate for multimodality treatment solely to reach LT. [4]

- Consider HAI, resection, ablation or other liver-directed options when they offer disease control or complete clearance. Before arterial intervention, document hepatic arterial anatomy and a possible future reconstruction strategy. Avoid a plan in which intentionally excessive treatment creates the liver failure used to justify LT. [6]
- OPEN proposed local monitoring: CT/MRI approximately every eight weeks during active therapy, CEA at oncology reviews, reassessment of functional/liver reserve and repeat two-surgeon HPB review after meaningful response and within 30 days before LT activation. PET quantification and ctDNA are not independent eligibility tests. [16] [17]
- Advance to final selection only if disease remains controlled, no extrahepatic/local recurrence is found, persistent unresectability is independently confirmed, medical fitness remains acceptable, and the chosen clinical or approved study stratum plus allocation route is documented.
- Hold for indeterminate lesions, NED without a clear LT rationale, discordant biomarkers, unresolved primary margins, unreviewed treatment toxicity or an unapproved expansion. Progression, confirmed extrahepatic/local recurrence or prohibitive medical risk stops the guidance-aligned route. [2]
- If a molecular subgroup is receiving ICI, retain the separate graft-risk assessment and drug chronology. Activity of immunotherapy in MSI-H CRC does not establish an LT indication or safe post-LT exposure. [18]

5 LISTING AND SURGICAL PROCEDURE

- NLRB CRLM guidance recommends consideration of MMaT minus 20 with a floor of 15. It is not a standardized exception, guaranteed score or the same priority intervention as TransMet. Keep the approved exception, calculated MELD and expected wait separately documented. [2] [1]
- Before recipient activation, document the intended DDLT or LDLT route, actual exception decision if applicable, expected waiting time, and a recipient-benefit assessment against contemporary non-LT care. Do not promise priority or treat DCD, an extended-criteria graft or machine perfusion as evidence that no other candidate could benefit. [1] [5]
- For LDLT, retain independent donor advocacy and donor-team assessment, freedom to withdraw, and separate consent covering uncertain recipient benefit, operative and long-term morbidity, recovery, financial burden and possible retransplantation. Do not relax donor safety criteria or make possession of a living donor a clinical eligibility requirement. [1] [5] [19]
- The living-donor literature reports low but nonzero mortality and incomplete recovery follow-up. Use current center- and graft-specific counseling; do not pool overlapping registry studies or portray donor survival as proof of complete recovery. [19]

ALLOC-B1 — WHAT ARE WE APPROVING?

Evidence and uncertainty before the shared rule

The separation is a governance requirement, not an empirical survival cutoff. A clinically appropriate patient can lack exception eligibility, and an approved exception does not prove clinical benefit. LDLT requires an independent donor determination; a registry entry is not permission to perform experimental transplantation. [1] [2] [20] [21]

Evidence-to-threshold distinction: Six domains: local checklist architecture, not six mandatory national approvals.; 100% applicable completion: governance safeguard, not an outcome-derived percentage.; Hold pending required authorization: operational implementation of separate clinical, donor and research responsibilities.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require 100% completion of the applicable clinical, allocation, donor and research determinations before activation. [2] [1] [20] [21]

- Complete six separate dated determinations: clinical suitability; standardized-exception eligibility; nonstandard NLRB review status; calculated-MELD access; LDLT donor clearance; and research authorization.
  - Mark nonapplicable domains explicitly with a reason—do not manufacture six approvals for every patient.
- Any required missing, pending or adverse determination is a HOLD.
  - An exception application is not approval, a donor is not recipient benefit, and clinical approval is not study activation.
- Recheck the current policy/guidance version before each request; record actual decision and priority rather than predicted entitlement.

Why retain or change it: Retain 100% completion of applicable determinations with dated responsible owners and reasoned N/A entries. A required unresolved determination means HOLD, not an assumed refusal or an authorization by default.

INFORMATION REQUIRED BEFORE THE DECISION

Signed suitability record; exact allocation route and policy version; exception outcome; donor-team status; study and institutional activation record.

Clinical versus allocation: Policy 9.4 permits discretionary requests where appropriate; absence of favorable diagnosis-specific guidance is not absence of a review mechanism, and a request does not promise approval.

Roadmap / protocol reconciliation: Use the same clinical-versus-allocation separation in every standard and high-risk pathway.

Meeting record: ALLOC-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B2 — WHAT IS THE RELEVANT ORGAN OPPORTUNITY COST?

Evidence and uncertainty before the shared rule

Use pathway-level benefit from an agreed entry date, not recipient-only survival. Compare with contemporary tumor-specific care and actual local access. LDLT avoids one direct deceased-donor allocation but creates healthy-donor risk and may generate rescue graft use. Graft quality/perfusion changes do not make opportunity cost zero. [21] [1] [22] [19] [23] [3] [24]

Evidence-to-threshold distinction: Two reviewers: local independence safeguard, not validated optimum.; 6/12/18 months: sensitivity scenarios for access/dropout, not mandatory waits.; HCC ≥60% five-year ITT OS: one expert trial-design benchmark; not a cross-cancer allocation policy.; All grafts: count initial and rescue/retransplant grafts; donor morbidity remains a separate outcome.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require two independent recipient-benefit reviews and an explicit organ-use case; do not adopt a universal five-year survival cutoff. [3] [19] [21] [1] [22] [23] [24]

- Proposed gate: two reviewers not both responsible for recruitment must document expected incremental benefit over the actual contemporary alternative, with unresolved disagreement referred to the full committee.
- For DDLT, show local expected access/dropout and sensitivity scenarios at 6, 12 and 18 months, total graft use including retransplantation, and plausible alternative recipients.
  - These are planning scenarios, not imposed waiting periods.
- For LDLT, separately show healthy-donor risk and potential downstream deceased-donor graft use.
  - Do not call DCD, extended-criteria grafts or machine perfusion "zero opportunity cost." No priority changes follow from an AI-selected survival threshold.

Why retain or change it: Retain independent benefit and organ-use reviews, but require disease-specific ITT evidence and explicit uncertainty. No universal five-year survival cutoff or numerical allocation weight is adopted.

INFORMATION REQUIRED BEFORE THE DECISION

Same-eligibility/time-zero comparator; expected access; dropout; donor/graft type; uncertainty intervals; graft loss/retransplant estimate; written independent reviews.

Clinical versus allocation: Only the applicable allocation process determines priority. Favorable post-LT prognosis, evidence grade and urgency are not interchangeable measures.

Roadmap / protocol reconciliation: Compare against the cancer roadmap's actual resection, ablation, HAI, radiation or systemic alternative—not historical untreated survival.

Meeting record: ALLOC-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B3 — WHAT MUST DONOR CONSENT AND FOLLOW-UP INCLUDE?

Evidence and uncertainty before the shared rule

Donor mortality is low but not zero; complication, recovery and financial burdens are more frequent. Registry missingness and selected healthy donors limit reassurance. Consent must include uncertain recipient benefit, recurrence and possible early graft loss, without pressure to proceed or relax donor criteria. [19] [25] [26] [27] [1]

Quantitative donor context: Kelly reports 8/8,619 deaths by 90 days across eras; A2ALL recorded ≥1 complication in 296/740 donors. These are different outcomes/eras, not poolable rates. The complete donor evidence and follow-up limitations are in ALLOC-B3.

Evidence-to-threshold distinction: Baseline/6 weeks/6 months/1/2 years then annually to five: proposed support schedule; current mandatory reporting remains separately checked.; One sentinel event: conservative review/pause trigger, not a statistically estimated harm boundary.; No relaxed donor criteria: protection against shifting uncertain recipient benefit onto a healthy person.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Do not relax donor criteria; fund independent follow-up and pause after a serious sentinel donor event. [1] [19] [25] [26] [27]

- Proposed readiness gate: independent donor evaluation/advocacy, documented freedom to withdraw, recipient-benefit uncertainty consent, and funded medical, psychological, recovery and financial follow-up before investigational LDLT.
- Proposed local follow-up: baseline, six weeks, six months, one year, two years, then annually through five years.
  - Meet current OPTN reporting duties separately; this five-year support plan is not claimed to be a national reporting requirement.
- One donor death, life-threatening complication, permanent major disability or liver failure requiring transplantation triggers an immediate pause of the relevant investigational LDLT activity and independent review.
  - Other serious donor harm also warrants review; reopening needs independent authorization.

Why retain or change it: Retain independent advocacy, ability to withdraw, funded medical/psychological/financial follow-up and sentinel-event pause. Add explicit denominator/era caveats to consent estimates and collect local outcomes prospectively.

INFORMATION REQUIRED BEFORE THE DECISION

Graft/resection-specific and center-specific risk; donor recovery/employment baseline; independent consent and advocate records; funded follow-up/support; sentinel-event response owner.

Clinical versus allocation: LDLT clearance is separate from recipient selection; the potential need for retransplantation must be included in organ-use discussions.

Roadmap / protocol reconciliation: The same independent donor safeguards apply whether recipient disease is standard or experimental.

Meeting record: ALLOC-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

- Before LT, reconfirm staging and inspect for unexpected extrahepatic disease. Obtain pathology of suspicious findings; malignant extrahepatic disease stops this protocol operation. Complete native hepatectomy and map viable and nonviable sites on explant.
- Living-donor RAPID or another auxiliary/two-stage operation is not a default alternative when a whole graft is unavailable. It requires a separate surgical study, graft-growth/functional criteria, small-for-size and interval-progression safeguards, and independent donor assessment. The old 2–3-week completion schedule is not an automatic operative order.

6 IMMUNOSUPPRESSION AND ADJUVANT THERAPY

- Use an approved transplant regimen with individual renal, infection, rejection and oncologic assessment. mTOR conversion and tacrolimus minimization may be considered after appropriate healing and vascular assessment, but no high-risk CRLM recurrence-prevention regimen or universal trough target is established. [4] [5]
- No routine post-LT adjuvant chemotherapy standard is adopted here. Any such plan must specify regimen, recovery requirements, duration and graft/marrow/interaction monitoring in an approved study

or individualized MDT decision. Four-week ctDNA status must not alone start, stop or extend chemotherapy. [17]

7 POST TRANSPLANT SURVEILLANCE

- Proposed local schedule for approval: chest/abdomen/pelvis imaging and CEA every three months for years one to three, then every six months through year five, with subsequent follow-up individualized. Choose liver MRI where appropriate; assess kidney function and cumulative imaging burden. [4]
- Use PET for an unresolved clinical or imaging question, not automatically every six months. Obtain colonoscopic surveillance through the colorectal team based on prior examination and resection history.
- ctDNA remains a research/risk adjunct. Confirm a new positive result and perform directed restaging/MDT review; do not initiate empiric therapy with negative imaging solely because ctDNA is detected. [17]

8 RECURRENCE MANAGEMENT

- Restage and discuss each recurrence in the colorectal/transplant MDT. Selected limited pulmonary or other recurrences may be treated with resection, ablation or radiation; systemic treatment depends on disease burden, biology, prior exposure, graft function and patient priorities. [4] [3]
- Record recurrence-treatment morbidity, further recurrence, time without treatment and survival after recurrence. Local salvage does not imply cure or justify accepting uncontrolled disease before LT. [3]

CRLM-B4 — HOW MUCH RECURRENCE/GRAFT USE IS ACCEPTABLE FOR THE OBSERVED SURVIVAL BENEFIT?

QUESTION AND COMPARISON

How should expected survival, recurrence treatment, graft use and non-LT alternatives determine whether a high-risk CRLM transplant strategy is worthwhile?

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TransMet 2024 [3] Multicentre open-label randomized trial, 20 European centres, 157 screened and 94 randomized (47 LT-plus-chemotherapy, 47 chemotherapy); 63 screened excluded, including 13 with equivocal unresectability and 36 with progression. Primary source | Full-text PDF Locator: Lancet 2024;404:1107–1118. PDF pp.2–4 eligibility/committee; pp.5–6 trial flow/Table 1; pp.8–9 recurrence, safety, retransplantation, Figure 3; p.10 QoL/limitations. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Primary time origin was randomization. Median follow-up 59.3 months. Five-year ITT OS was 56.6% versus 12.6% (HR 0.37, 95% CI 0.21–0.65); per-protocol OS 73.3% versus 9.3%. Of 47 assigned LT, 38 received LT and 36 were per-protocol; nine had progression-related exclusion/dropout. The control arm included seven subsequent resections and two LT crossovers. Safety population comprised 40 grafted patients. Among 36 per-protocol LT recipients, 26 recurred; 19/26 received chemotherapy and 12/26 local curative-intent treatment. Median per-protocol PFS was 17.4 versus 6.4 months, and secondary PFS after salvage was 35.4 months in the LT arm. Three of 36 per-protocol recipients required retransplantation (primary nonfunction, donor gallbladder cancer, caval obstruction/large-for-size); one postoperative death was reported. Serious adverse events occurred in 32/40 safety LT recipients. Median wait to LT was 50.5 days, which limits transfer to longer-wait systems. **Do not infer:** Strong evidence for the selected trial population, not for BRAF-mutant, ECOG 2, active extrahepatic, truly NED, resectable or broadly expanded disease. PP survival conditions on reaching treatment and must not replace ITT. The Results recurrence denominator is 26/36; Figure 3 caption says 28 recurrences, consistent with a broader all-grafted report, so do not mix these denominators. QoL collection had only 55/74 per-protocol baseline questionnaires and is descriptive, not proof of a large QoL gain. Priority allocation and rapid access were part of the treatment strategy.
Dueland 2026 · SECA-II ten-year [8] Prospective SECA-II extended follow-up, 25 recipients from April 2012 to November 2023; analysis cutoff 1 January 2025. Includes the original SECA-II cohort. Primary source | Full-text PDF Locator: Ann Surg 2026 early report DOI 10.1097/SLA.0000 000 000 007 200; PDF p.2 methods/time origins and early-death handling; p.3 Table 1 and outcomes; pp.4–5 salvage, late death, cure-count discrepancy and complications. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Time zero LT; median follow-up of survivors 97.2 months. Median OS 128.2 months (95% CI 74.8–181.6); five- and ten-year OS 77.8% and 51.5%. Median RFS 24.3 months; 12/25 relapsed, six initially lung-only. Median post-recurrence survival 62.9 months (25.7–101.1); nine had surgery as first recurrence treatment, but only three were classified curative-intent at that first treatment. Twelve had no recurrence over median 74 months; four exceeded ten years without recurrence. Four patients died of metastases more than five years after LT. Reported serious complications: five grade IIIa, three IIIb, one grade V hepatic-artery-thrombosis death. **Do not infer:** No contemporaneous control and only 25 recipients; highly favorable biology cannot justify broad expansion. Abstract speculates up to 15 potentially cured, body p.5 says up to 13; neither is an observed, validated cure fraction. Use observed survival/recurrence instead. The discussion's assertion that RFS is not valid for efficacy should not erase recurrence burden, treatment harms or QoL; RFS is insufficient alone, not unimportant. Updated survival supersedes rather than adds an independent 83% SECA-II estimate.

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SECA-II arm D [14] Prospective SECA-II arm D expanded-criteria cohort, ten recipients, without control. Primary source | Full-text PDF Locator: BJS Open 2020;4:467–477; full text methods, Tables 1–3 and survival figures. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** From LT, median DFS four months and median OS 18 months. Rapid recurrence and limited survival contrast with strictly selected SECA-II, despite use of donor grafts intended to broaden access. **Do not infer:** Very small and uncontrolled; simultaneous biologic, anatomic and donor expansions prevent estimating the independent effect of prior pulmonary disease, BRAF status, grade or donor quality. Negative experience argues against opening several expansions at once, not that every single deviation has proven futility.
Chandra 2026 · HAIP comparator [6] Single-centre retrospective HAIP cohort: 1,110 patients treated in 2006–2016, 483 initially unresectable after exclusions; 83 met modified transplant criteria, but only 23 were pretreated, chemotherapy-controlled transplant-like candidates. Primary source | Full-text PDF Locator: Complete accepted manuscript, PDF pp.6–11: cohort derivation, modified criteria, conversion, survival and adverse events; subsequent tables. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** Time zero was HAIP placement, not a common LT-versus-HAIP eligibility date. In the 23 selected pretreated patients, 18/23 (78%) converted to resection; only 10/18 were R0, and 17/18 recurred, with salvage treatment in 15. Median follow-up was 97.7 months; median OS 61 months (95% CI 36–92), five-year OS 53% (36–79), PFS 13 months (9.6–22), hepatic PFS 16 months (11–61). Median treatment comprised five HAIP cycles and nine systemic cycles. Biliary sclerosis requiring stents occurred in 3/23 (13%); one pump-arterial dissection required revision. **Do not infer:** Selection from a large denominator, modified eligibility, expert-centre availability, retrospective ascertainment and a different time origin prohibit equivalence claims against TransMet. The high conversion rate is clinically important counterevidence to declaring unresectability before specialist conversion review. It does not make HAIP compulsory or establish that its toxicities are an indication to create for LT.

Study • population • source	Results • interpretation boundary
Wehrle 2024 · intensive LRT [13] Single-centre intensive pre-LT locoregional-treatment protocol: 27 evaluated, 16 included, nine transplanted (seven living donor). Abstract-only verification. Primary source | TMC: retrieve PDF Locator: Liver Transpl 2024;30:1238–1249; PMID38 833 301 published abstract; complete methods pending. Review depth: Published primary abstract reviewed; complete PDF not available in searched depots/Downloads	**Reported result:** Median follow-up 700 days; three recurrences. Reported RFS one year 6/8 (75%) and two years 3/5 (60%) reflects very small mature denominators. **Do not infer:** This is actual published experience with NED/injury routes, so it must not be described as nonexistent; it is not controlled proof of benefit or a universal cycle-count requirement. Patients likely overlap occult-explant/PET/ctDNA cohorts. Do not intensify treatment to manufacture liver failure.
NLRB oncology guidance [2] Official Adult MELD Exceptions for Transplant Oncology guidance, December 2025 posting checked 24 October 2026. Allocation-review guidance, not national clinical policy or proof of efficacy. Primary source | Full-text PDF Locator: Official PDF NET pp.8–9; CRLM pp.9–10; introduction on purpose/status. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** No efficacy denominator. NET guidance has no explicit numerical age or hepatic-volume cutoff. CRLM wording excludes >10% diameter growth or new lesions and labels this RECIST 1.1, although conventional RECIST 1.1 progression generally uses at least 20% summed-diameter increase plus an absolute change; record both, do not silently equate them. **Do not infer:** Clinical judgment, exception endorsement and graft availability are distinct. A reasonable research hypothesis or living donor does not override allocation guidance or demonstrate acceptable benefit. At exactly CEA 80 the initial/extension wording differs; the draft's strict <80 harmonization is a conservative local rule.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

The relevant endpoint is the entire strategy from a shared eligibility/randomization date: access to a first graft, wait-list progression, perioperative death, graft failure, recurrence, salvage treatment, long-term survival and QoL. Recurrence does not erase all benefit, but is a substantial burden; SECA-II shows both long post-recurrence survival and late metastatic death. TransMet's 56.6% ITT survival is the appropriate randomized headline, with 73.3% PP provided only secondarily. [3] [8] [14] [6] [13] [2]

- TransMet's nine progression dropouts and three retransplants show why post-LT survival alone is inadequate. The 2026 SECA-II ten-year OS was 51.5%, not its five-year 77.8%; four metastatic deaths occurred after five years. Arm D's 18-month median OS cautions against assuming every expanded candidate obtains sufficient benefit. Selected HAIP/conversion patients can have durable survival without a graft, but cross-study comparisons cannot calculate an individual's LT gain.

Certainty / applicability: Moderate-to-high for a survival advantage in selected TransMet patients; low for an individual numerical benefit model, long-wait extrapolation, high-risk graft strategies or predetermined retransplant policy. Long-term SECA-II is informative but small and uncontrolled.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- No validated minimum RFS, individual five-year benefit percentage or fixed number of salvage operations establishes acceptable LT value. The two-year/three-year thresholds sometimes used in economic models are not recipient-selection validation. One planned graft and safety-review triggers are institutional ethical/operational proposals.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
TransMet 2024 [3] "three (8%) of 36 patients who received a transplant were retransplanted" Published PDF p9 / journal p1115, Results Source | Full-text PDF	Additional graft use is real. The 36 per-protocol recipients differ from the 40-recipient safety population.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Accept that recurrence may coexist with meaningful survival benefit, but plan one graft and account for recurrence treatment and all additional graft use. [3] [8] [14] [6] [13] [2]

- Proposed activation gate: a patient-specific benefit comparison against accessible non-LT care, feasible recurrence-treatment options, and a realistic local wait/treatment plan.
  - Reassess about every eight weeks while waiting; no unsupported maximum-wait exclusion is imposed.
- Plan one graft per recipient; no planned retransplantation for oncologic recurrence.
  - Rescue of early nononcologic graft failure remains a separate standard transplant/allocation decision and is never promised or withheld by this rule.
- No low recurrence-rate/RFS cutoff is substituted for survival or quality-of-life benefit.
  - Record all-screened outcomes, recurrence treatment burden, time off therapy, graft loss/retransplantation and functional recovery.
  - Apply the one-event early safety pause in ALLOC-B4 to newly activated experimental cohorts.

Why retain or change it: Keep OPEN a one-planned-graft strategy, with no planned oncologic retransplantation. Emergency graft rescue must be an independent transplant/allocation decision rather than automatic denial or promise. Before activation document the feasible non-LT sequence, expected wait and bridging plan, anticipated recurrence salvage options, recipient/graft risk, and donor-specific burden where applicable. Track all screened/enrolled patients from common eligibility, not only recipients; report ITT-like survival, recurrence sites/treatment, grafts per patient, donor outcomes and QoL. A safety event triggers review under the separate allocation governance plan; it is not a literature-derived stopping boundary. Decision remains blank.

INFORMATION REQUIRED BEFORE THE DECISION

Contemporary comparator; current fitness/control; recurrence options; local access estimate; patient priorities; graft-use and follow-up plan.

Clinical versus allocation: Do not promise trial survival where waiting/access differ materially. Count all grafts and downstream allocation consequences, including LDLT recipients who later need DDLT.

Roadmap / protocol reconciliation: Use the CRLM roadmap for recurrence and non-LT alternatives; compare real local options rather than historical chemotherapy alone.

Meeting record: CRLM-B4 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

9 DATA COLLECTION AND OUTCOMES

- Register every referred or screened candidate in the institutionally approved system, not only recipients. Record acceptance or rejection, treatment entry, response, listing, exception decision, conversion to resection or ablation, progression, toxicity, decompensation, death, patient withdrawal and access-related loss.
- Report overall survival from the prespecified entry point and separately from listing and transplantation. Keep recurrence-free survival, recurrence incidence, graft survival and survival after recurrence distinct, with definitions, confidence intervals, numbers at risk and follow-up completeness. Do not call recipient-only survival an intention-to-treat outcome.
- Capture 90-day recipient morbidity and mortality, biopsy-defined rejection, infection, arterial and biliary complications, graft loss, retransplantation and total grafts used. Record recurrence site, salvage treatment, time off systemic therapy, functional recovery and quality of life, not survival alone.
- Before a transplanting study opens, biostatistics and the independent safety board must approve indication-specific monitoring, evaluable denominators, pause authority and reopening criteria. ALLOC-B4 now proposes explicit initial-cohort review triggers for ratification; they are not validated efficacy/futility boundaries. No universal cross-cancer five-year survival cutoff is adopted.
- Use a contemporary comparator with the same eligibility and time zero. Account for the interval needed to survive to transplantation, crossover, competing risks and overlapping study cohorts. A nonsignificant difference is not proof of equivalence.

ALLOC-B4 — WHAT DATA ARE REQUIRED BEFORE EXPANSION?

Evidence and uncertainty before the shared rule

A small pilot can test feasibility, staging fidelity, data capture and acute safety, not prove durable survival benefit. Define time zero, all-screened enrollment, waitlist failures, post-LT survival/recurrence, competing death, all grafts, donor harms and patient-reported outcomes before opening. An independent board needs actual pause/reopening authority. [22] [28] [20] [19] [29] [30]

Zero observed events among ten fully followed independent patients still has a one-sided exact 95% binomial upper bound of 25.9% (1−0.05^(1/10)). Thus ten survivors cannot establish a <5% event risk. Ninety percent data completeness permits one missing patient; safety events and vital status require active ascertainment, not tolerance of unaccounted loss.

Evidence-to-threshold distinction: Ten per stratum and interim after five at 90 days: local feasibility cap and safety review cadence.; Twelve-month minimum: earliest expansion discussion, not proof of long-term cancer benefit.; ≥90%: proposed non-vital key-measure completeness threshold; not permission to ignore deaths/graft events or unknown status.; One early death/graft loss/re-LT: review/pause trigger, not automatic futility or withholding rescue.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use a small staged pilot with all-screened accounting and explicit safety pauses; no automatic expansion on favorable recipient survival. [22] [28] [20] [19] [29] [30]

- Proposed initial cap: ten transplants per separately approved biological stratum.
  - Review after the first five have 90-day follow-up; no expansion decision until all first ten have at least 12-month follow-up.
  - Reconcile vital status, graft events and known serious harms for every participant; require ≥90% completeness for other prespecified key outcomes.
- In an initial experimental cohort, one all-cause recipient death, graft loss or retransplantation within 90 days triggers a pause of further experimental enrollment/transplantation pending independent review.
  - A later possibly treatment-related death/graft loss or a donor sentinel event also triggers review/pause.
  - Rescue care is never withheld; reopening requires independent authorization.
- These are proposed conservative review triggers, not statistically validated efficacy/futility boundaries.
  - Before opening, biostatistics and the independent board must ratify or replace them with indication-specific rules, denominators, pause authority and reopening criteria.
- Capture every screened candidate, including nonrecipients, conversion to resection, dropout, deaths and access barriers; report survival from entry and from LT, recurrence, graft use, donor recovery and quality of life.
  - Ten survivors alone do not establish benefit or authorize expansion.

Why retain or change it: Retain the small staged pilot as an OPEN governance proposal, not an efficacy claim. Clarify that known safety/vital-status data must be reconciled for every enrolled patient; ≥90% applies only to other prespecified key measures. Indication-specific statistical design may replace the proposed numbers before opening.

INFORMATION REQUIRED BEFORE THE DECISION

Locked eligibility strata; prospective entry/time-zero dataset; modern comparator; adverse-event definitions; follow-up-completeness report; independently signed statistical and safety plan.

Clinical versus allocation: Study monitoring never changes exception criteria or guarantees future access. Audit access by referral, geography, resources and donor availability without turning protected characteristics into exclusion rules.

Roadmap / protocol reconciliation: Retain cancer-roadmap treatment alternatives and document why non-LT care was chosen when candidates do not proceed.

Meeting record: ALLOC-B4 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

10 CORRELATIVE STUDIES AND UNRESOLVED DECISIONS

- Keep ctDNA, PET metabolic volume, explant genomics and pharmacokinetics in prespecified research modules. Record assay/platform, timing and missingness; a cohort-derived cutoff is a hypothesis until externally validated. [17] [16]

Resolve CRLM-B1–B4 and ALLOC-B1–B4. The prior A-versus-B comparison table was removed because it contradicted the main text and treated unvalidated rules as accepted practice. Decisions require exact final wording, responsible owners, date and approval status.

Evidence review record: Five analyst comments and reconciliation | Prospective outcomes dataset

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REVISION NOTE

Revised 2026-10-24 06:55 CDT. Evidence-before-recommendation review of all 27 B questions. Earlier protocol versions and external analyst records are preserved; this is an Analyst A synthesis, not a new external-analyst vote.

Problem: The earlier B proposals had an uneven evidence base and emphasized a stance before its supporting and contrary data. **Correction and reason:** Added question-level primary-study tables with selection, denominators, outcome clocks, harms, limitations, exact source excerpts and threshold provenance before each OPEN proposal. Sources/PDF links are adjacent; missing full texts remain flagged.

Problem: Local operational numbers could be mistaken for validated eligibility or allocation thresholds. **Correction and reason:** Separated outcome-derived estimates, guideline/policy wording and local safeguards; revised proposals when the deeper review changed their evidentiary basis. Institutional decisions remain blank.

Approval record Clinical lead _____ Allocation review _____ Research authorization if required _____ Donor review if applicable _____ Date _____